ECO101
CHAPTER-22
PERFECT COMPETITION

Every firm shares two things with all other firms. First, every firm has to answer certain questions: (1) what price should the firm charge for the good it produces and sells? (2) How many units of the good should the firm produce? (3) How much of the resources that the firm needs to produce its good should it buy? All firms must answer all three of these questions. Second, every firm finds itself operating in a certain market structure. A market structure is a firm's environment or setting whose characteristics influence the firm's pricing and output decisions.

THE THEORY OF PERFECT COMPETITION

Properties of perfect competition:

1. **There are many sellers and many buyers, none of which is large in relation to total sales or purchases.**

This assumption speaks to both demand (the number of buyers) and supply (the number of sellers). Given many buyers and sellers, each buyer and each seller may act independently of other buyers and sellers, respectively, and each is such a small part of the market as to have no influence on price.

2. **Each firm produces and sells a homogeneous product.**

Each firm sells a product that is indistinguishable from all other firms' products in a given industry. As a consequence buyers are indifferent to the sellers.

3. **Buyers and sellers have all relevant information about prices, product quality, sources of supply, and so forth.**

Buyers and sellers know who is selling what, at what prices, at what quality, and on what terms. In short, they know everything that relates to buying, producing, and selling the product.

4. **Firms have easy entry and exit.**

New firms can enter the market easily, and existing firms can exit the market easily. There are no barriers to entry or exit.

A Perfectly Competitive Firm Is a Price Taker

A perfectly competitive firm is a price taker, which is a seller that does not have the ability to control the price of its product; in other words, such a firm “takes” the price determined in the market. A firm is a price taker because it finds itself one among many firms whose supply is small relative to the total market supply.

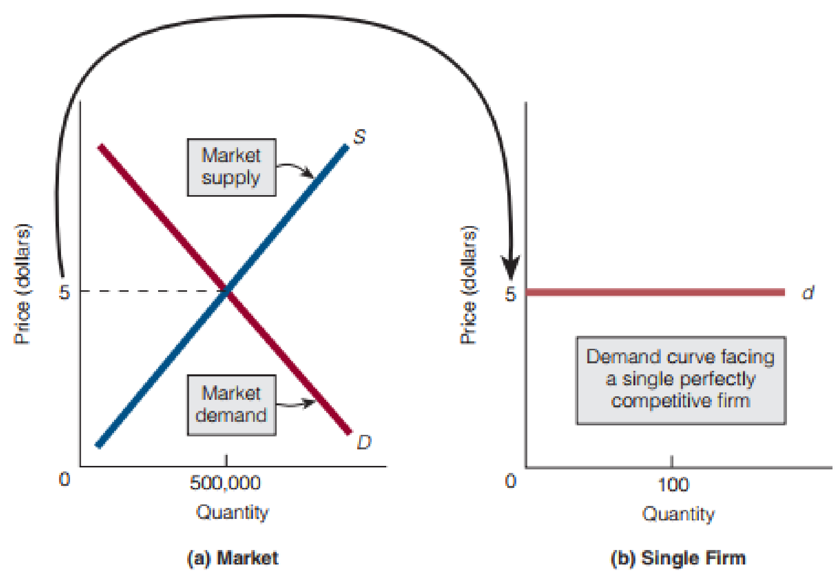
The Demand Curve for a Perfectly Competitive Firm Is Horizontal

The perfectly competitive setting has many sellers and many buyers. Together, all buyers make up the market demand curve; together, all sellers make up the market supply curve. An equilibrium price is established at the intersection of the market demand and market supply curves [Exhibit 1(a)].

EXHIBIT 1

The Market Demand Curve and Firm Demand Curve in Perfect Competition

(a) The market, composed of all buyers and sellers, establishes the equilibrium price. (b) A single perfectly competitive firm then faces a horizontal (flat, perfectly elastic) demand curve. We conclude that the firm is a price taker: it takes the equilibrium price established by the market and sells any and all quantities of output at this price. (The capital *D* represents the market demand curve; the lowercase *d* represents the single firm’s demand curve.)



When the equilibrium price has been established, a single perfectly competitive firm faces a horizontal (flat, perfectly elastic) demand curve at the equilibrium price [Exhibit 1(b)]. In short, the firm takes the equilibrium price as given—hence the firm is a price taker—and sells all quantities of output at this price.

The Marginal Revenue Curve of a Perfectly Competitive Firm Is the Same as Its Demand Curve

Total revenue is the price of a good multiplied by the quantity sold. If the equilibrium price is \$5, as in Exhibit 2(a), and the perfectly competitive firm sells 3 units of its good, its total revenue is \$15. If the firm sells an additional unit, bringing the total number of units sold to 4, its total revenue is \$20. A firm's marginal revenue (MR) is the change in total revenue (TR) that results from selling one additional unit of output (Q):

$$MR = \frac{\Delta TR}{\Delta Q}$$

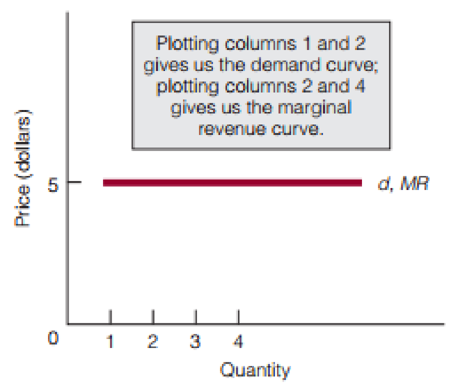
EXHIBIT 2

The Demand Curve and the Marginal Revenue Curve for a Perfectively Competitive Firm

(a) By computing marginal revenue, we find that it is equal to price. (b) By plotting columns 1 and 2, we obtain the firm's demand curve; by plotting columns 2 and 4, we obtain the firm's marginal revenue curve. The two curves are the same.

(1) Price	(2) Quantity	(3) Total Revenue = (1) × (2)	(4) Marginal Revenue = $\Delta TR/\Delta Q$ = $\Delta(3)/\Delta(2)$
\$5	1	\$5	\$5
5	2	10	5
5	3	15	5
5	4	20	5

(a)



(b)

Column 4 in Exhibit 2(a) shows that the firm's marginal revenue (\$5) at any output level is always equal to the equilibrium price (\$5). For a perfectly competitive firm, therefore, price (P) is equal to marginal revenue: For a perfectly competitive firm, **P = MR**

But if price is equal to marginal revenue, then the marginal revenue curve for the perfectly competitive firm is the same as its demand curve. A demand curve plots price against quantity, whereas a marginal revenue curve plots marginal revenue against quantity. If price equals marginal revenue, then the demand curve and marginal revenue curve are the same [Exhibit 2(b)]: For a perfectly competitive firm, Demand curve = Marginal revenue curve.

PERFECT COMPETITION IN THE SHORT RUN

For the perfectly competitive firm, a price taker, price is equal to marginal revenue (P= MR), and therefore the firm's demand curve is the same as its marginal revenue curve.

What Level of Output Does the Profit-Maximizing Firm Produces in the Short-run?

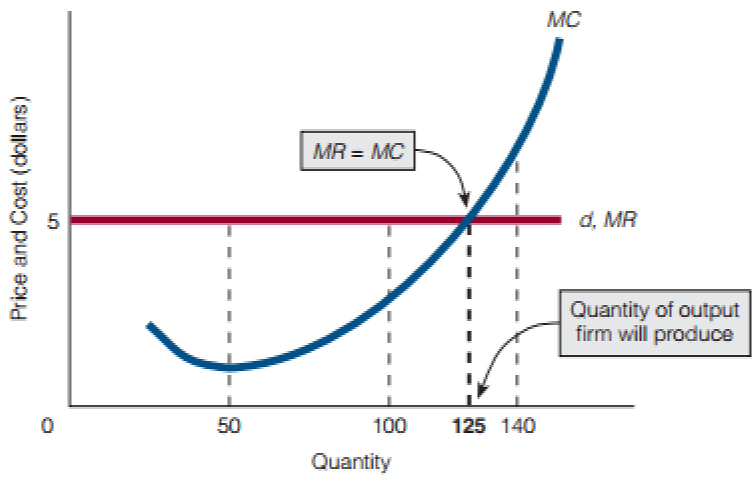
A competitive is a price taker as it takes the price determined by market demand and supply. Therefore the firm decides how much to produce to maximize profit. **Profit Maximization Rule** tells us that **p**rofit is maximized by producing the quantity of output at which MR = MC.

In Exhibit 3, the perfectly competitive firm's demand curve (d) and marginal revenue curve (MR, which is the same as d) are drawn at the equilibrium price of $5. The firm's marginal cost curve (MC) is also shown. On the basis of these curves, the firm will continue to increase its quantity of output as long as marginal revenue is greater than marginal cost. It will not produce units of output for which marginal revenue is less than marginal cost. Therefore, the firm will stop increasing its quantity of output when marginal revenue and marginal cost are equal. The profit maximization rule for a firm says, produce the quantity of output at which MR = MC. In Exhibit 3, MR = MC 5 at 125 units of output. For the perfectly competitive firm, the profit maximization rule can be written as P = MC because, for that firm, P =M R and MR = MC. Therefore, in perfect competition, profit is maximized when, P= MR = MC.

EXHIBIT 3

The Quantity of Output That the Perfectly Competitive Firm Will Produce

The firm's demand curve is horizontal at the equilibrium price. Its demand curve is its marginal revenue curve. The firm produces that quantity of output at which *MR* = *MC*.



On the basis of above diagram, the firm maximizes profit at a price $5 and quantity 125 units for which

P = MR = MC.

The Perfectly Competitive Firm and Resource Allocative Efficiency

Resources (or inputs) are used to produce goods and services the resources used in production have an exchange value that is approximated by the price that people pay for the good.

Producing a good—any good—until price equals marginal cost ensures that all units of the good are produced that are of greater value to buyers than the alternative goods that might have been produced. In other words, a firm that produces the quantity of output at which price equals marginal cost (P = MC) is said to exhibit resource allocative efficiency.

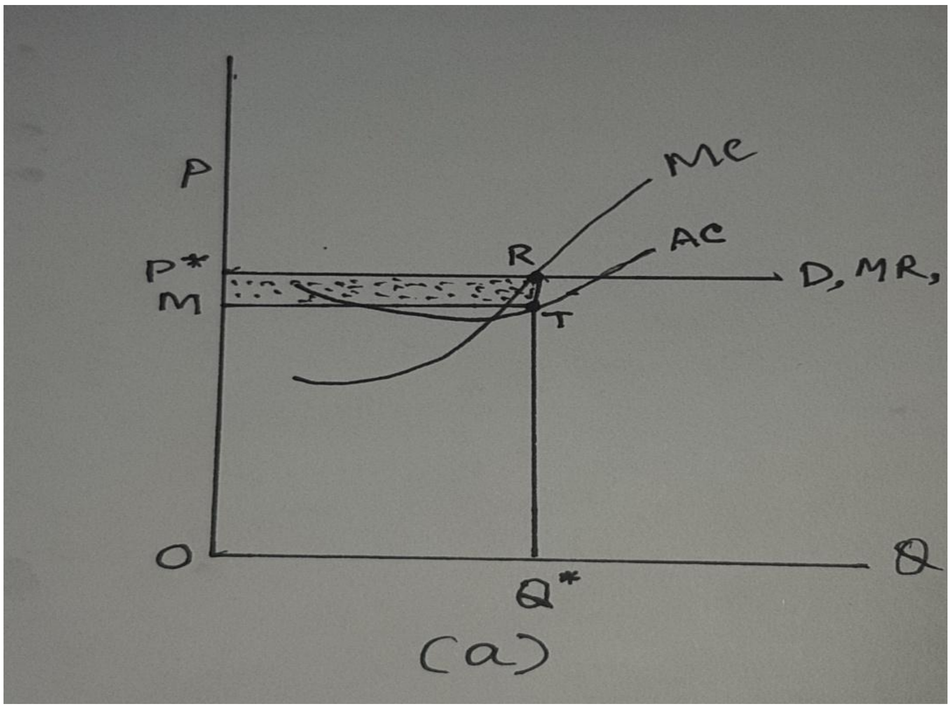
A competitive firm is allocative efficient in the short-run because: First, the perfectly competitive firm produces the quantity of output at which MR = MC. Second, for such a firm, P = MR. If the perfectly competitive firm produces the output at which MR =MC and if, for this firm, P = MR, then the firm produces the output at which P =MC . In short, the perfectly competitive firm is resource allocative efficient.

Short-run possible outcomes of a competitive firm

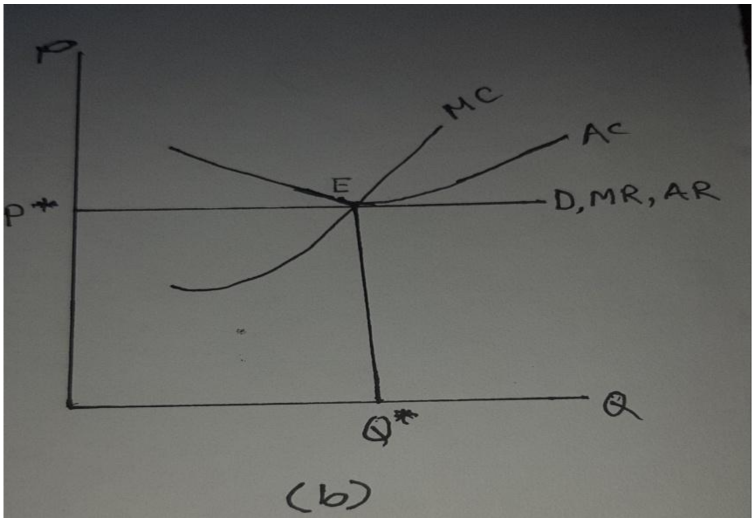
In the short-run a competitive firm has three possible outcomes:

(1) Firm can make economic profit

Perfect competitive firm in the short-run may make economic profit in terms per unit profit or total profit. At point R, MR = MC. Competitive firm produces Q* amount of output and sells at market determined price P*. Average revenue, AR = Q*R and average cost, AC = Q*T. Therefore, AR > AC. The firm makes economic profit per unit is RT. Total economic profit for Q* amount of output is MP*RT.



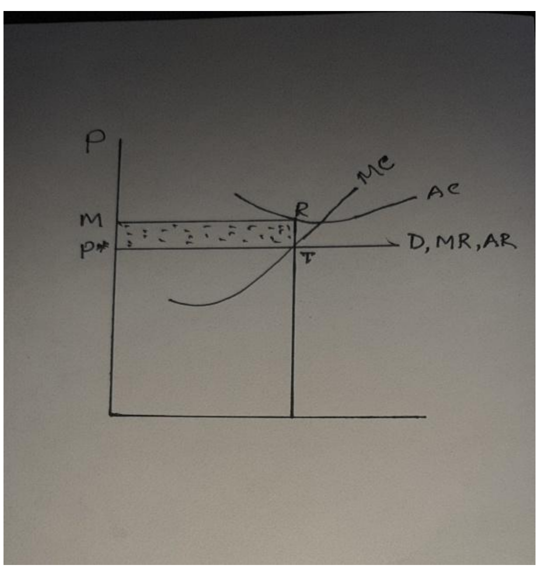
(2) Can make normal profit



Short-run competitive firm makes normal or zero economic profit if TR = TC or AR = AC. In figure (b) competitive firm produces Q* amount of output and sells at market determined price P*. At equilibrium E, MR = MC and also AR = AC. So that in this situation short-run competitive firm can make only normal or zero economic profit.

(3) Can incur economic loss

Short-run competitive firm may incur economic loss. Economic loss occurs if TR – TC < 0 or TR < TC or TC > TR or AR < AC OR AC > AR. In the following figure AC > AR by the amount RT which is per unit economic loss. Total economic loss = P*MRT.



Summary of Cases 1–3 *A perfectly competitive firm produces in the short run as long as price is above average variable cost (cases 1 and 3):*

$$P > AVC \rightarrow \text{Firm produces}$$

A perfectly competitive firm shuts down in the short run if price is less than average variable cost (case 2):

$$P < AVC \rightarrow \text{Firm shuts down}$$

We can summarize the same information in terms of total revenue and total variable costs. *A perfectly competitive firm produces in the short run as long as total revenue is greater than total variable costs (cases 1 and 3):*

$$TR > TVC \rightarrow \text{Firm produces}$$

A perfectly competitive firm shuts down in the short run if total revenue is less than total variable costs (case 2):

$$TR < TVC \rightarrow \text{Firm shuts down}$$

EXHIBIT 6

Q&A about Perfect Competition

Question	Answer
What four assumptions is the theory of perfect competition built on?	1. There are many buyers and many sellers. 2. Each firm produces and sells a homogeneous good. 3. Buyers and sellers have all relevant information about prices, product quality, sources of supply, and so forth. 4. Firms have easy entry into the market and easy exit out of the market.
What does it mean to say the perfectly competitive firm is a price taker?	The perfectly competitive firm *takes* the market-determined equilibrium price as the price at which it sells its product. The firm has no ability to control the price of the product it sells.
At what price does the perfectly competitive firm sell its product?	It sells at the price determined by the market. In other words, market demand and market supply determine the price of the good—say, \$10—and then the firm takes this price as the price at which it will sell its product.
What quantity does the single perfectly competitive firm produce?	The quantity at which $MR = MC$.
How do we know if the perfectly competitive firm is earning profit or incurring a loss?	If $P > ATC$ for the firm, then it is earning profit. If $P < ATC$ for the firm, then it is incurring a loss.
What is resource allocative efficiency, and is the perfectly competitive firm resource allocative efficient?	Resource allocative efficiency exists if firms produce the quantity of output at which $P = MC$. The perfectly competitive firm is resource allocative efficient. Proof: (1) The firm produces the quantity of output at which $MR = MC$. (2) In perfect competition, $P = MR$. (3) Because $P = MR$ and the firm produces the quantity at which $MR = MC$, it follows that $P = MC$. Hence, the firm is resource allocative efficient.

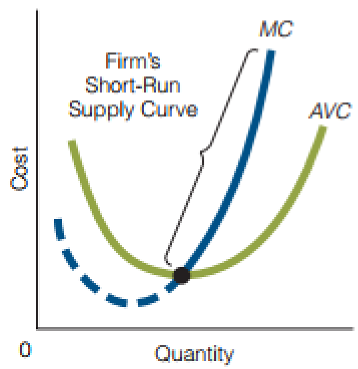
The Perfectly Competitive Firm's Short-Run Supply Curve

The perfectly competitive firm produces (supplies output) in the short run if price is above average variable cost. It shuts down (does not supply output) if price is below average variable cost. Therefore, the short-run (firm) supply curve is the portion of the firm's marginal cost curve that lies above the average variable cost curve. Only a price above average variable cost will induce the firm to supply output. The short-run supply curve of the perfectly competitive firm is illustrated in Exhibit 7.

EXHIBIT 7

The Perfectly Competitive Firm's Short-Run Supply Curve

The short-run supply curve is that portion of the firm's marginal cost curve that lies above the average variable cost curve.



If the perfectly competitive firm's short-run supply curve is the part of its marginal cost curve above its average variable cost curve, then deriving the short-run market (industry) supply curve is a simple matter: We horizontally add the short-run supply curves for all firms in the perfectly competitive market or industry.

Consider, for simplicity, an industry made up of three firms: A, B, and C. [See Ex

EXHIBIT 8

Deriving the Market (Industry) Supply Curve for a Perfectly Competitive Market In (a), we add (horizontally) the quantity supplied by each firm to derive the market supply curve. The market supply curve and the market demand curve are shown in (b). Together, they determine equilibrium price and quantity.

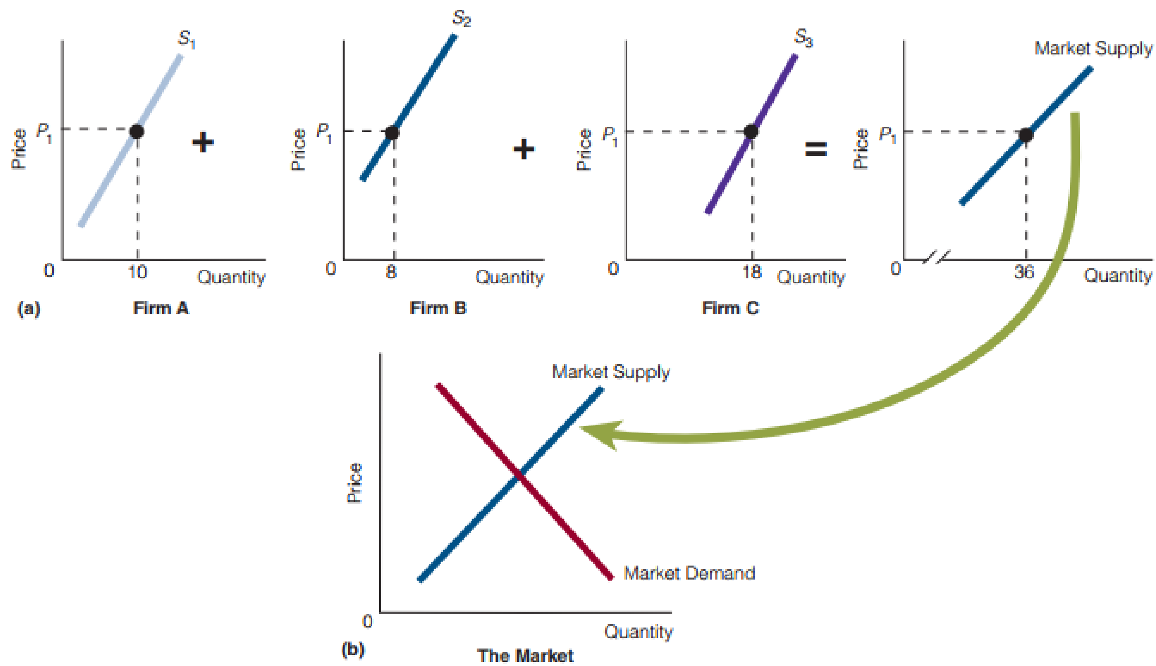


exhibit 8(a).] At a price P_1, firm A supplies 10 units, firm B supplies 8 units, and firm C supplies 18 units. One point on the market supply curve thus corresponds to P_1 on the price axis and 36 units (10, 8, 18 and 36) on the quantity axis. If we follow this procedure for all prices, we have the short-run market supply curve. This curve, shown in the market setting in part (b) of the exhibit, is used along with the market demand curve to determine equilibrium price and quantity.

Why Is the Market Supply Curve Upward Sloping?

When the demand and supply curves were introduced in Chapter 3, the supply curve was drawn upward sloping. To understand why, consider the following questions and answers: • Question 1: Why do we draw market supply curves upward sloping?

• Answer: Because market supply curves are the horizontal sum of firms' supply curves and firms' supply curves are upward sloping.

• Question 2: But why are firms' supply curves upward sloping?

• Answer: Because the supply curve for each firm is the portion of its marginal cost (MC) curve that is above its average variable cost (AVC) curve—and this portion of the MC curve is upward sloping.

• Question 3: But why do MC curves have an upward-sloping portion?
• Answer: According to the law of diminishing marginal returns, the marginal physical product (MPP) of a variable input eventually declines. When that happens, the MC curve begins to rise.

Conclusion: Because of the law of diminishing marginal returns, MC curves are upward sloping, and because MC curves are upward sloping, so are market supply curves.

PERFECT COMPETITION IN THE LONG RUN

The number of firms in a perfectly competitive market may not be the same in the short run as in the long run. For example, if the typical firm is making economic profits in the short run, new firms will be attracted to the industry and the number of firms will increase. If the typical firm is sustaining losses, some existing firms will exit the industry and the number of firms will decrease.

The following conditions characterize long-run competitive equilibrium:

1. Economic profit is zero; that is, price (P) is equal to short-run average total cost (SRATC): P = SRATC The logic of this condition is clear when we analyze what will happen if price is above or below short-run average total cost. If it is above, positive economic profits will attract firms to the industry in order to obtain the profits. If price is below, losses will result and some firms will want to exit the industry. Long-run competitive equilibrium cannot exist if firms have an incentive to enter or exit the industry in response to positive economic profits or losses. For long-run equilibrium to exist, there can be no incentive for firms to enter or exit. This condition is brought about by zero economic profit (normal profit), which is a consequence of the equilibrium price being equal to short-run average total cost.
2. Firms are producing the quantity of output at which price (P) is equal to marginal cost (MC):

$$P = M C$$

   Perfectly competitive firms naturally move toward the output level at which marginal revenue (or price, because, for a perfectly competitive firm, MR = P) equals marginal cost.
3. No firm has an incentive to change its plant size to produce its current output; that is, at the quantity of output at which P M 5 C, the following condition holds: SRATC > L RATC

To understand this condition, suppose SRATC > LRATC at the quantity of output established in condition

2. Then the firm has an incentive to change its plant size in the long run because it wants to produce its product with the plant size that will give it the lowest average total cost (unit cost). It will have no incentive to change its plant size when it is producing the quantity of output at which price equals marginal cost and SRATC equals LRATC.

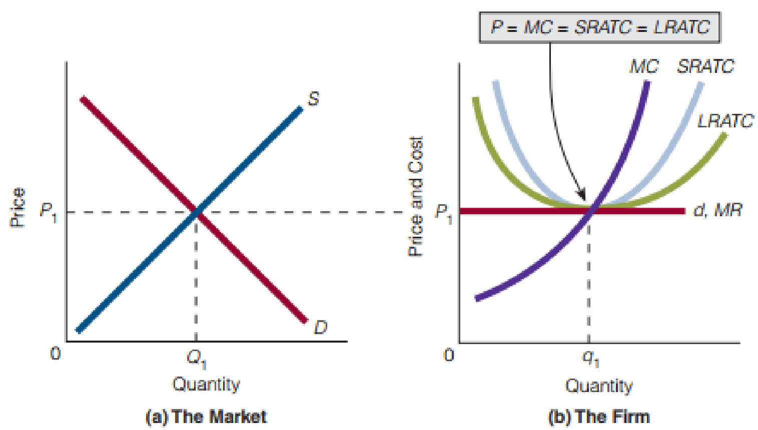
The three conditions necessary for long-run competitive equilibrium can be stated as shown in Exhibit 9: long-run competitive equilibrium exists when P = MC = SRATC = 5 LRATC . In conclusion, long-run competitive equilibrium exists when firms have no incentive to make any changes—that is, when there is no incentive for firms to do any of the following:

1. enter or exit the industry.
2. produce more or less output.
3. change their plant size.

EXHIBIT 9

Long-Run Competitive Equilibrium

(a) Equilibrium in the market. (b) Equilibrium for the firm. In (b), $P = MC$ (the firm has no incentive to move away from the quantity q_1 of output at which this equality occurs), $P = SRATC$ (there is no incentive for firms to enter or exit the industry), and $SRATC = LRATC$ (there is no incentive for the firm to change its plant size).



A firm that produces its output at the lowest possible per-unit cost (lowest ATC) is said to exhibit productive efficiency. The perfectly competitive firm is productively efficient in long-run equilibrium, as shown in Exhibit 9. Productive efficiency is desirable from society's standpoint because perfectly competitive firms are economizing on society's scarce resources and therefore not wasting them.

Sample questions for quiz

Define market structure
A market structure is a firm's environment or setting whose characteristics influence the firm's pricing and output decisions.

What are the properties of a competitive market?
(1) There are many sellers and many buyers, none of which is large in relation to total sales or purchases.
(2) Each firm produces and sells a homogeneous product.
(3) Buyers and sellers have all relevant information about prices, product quality, sources of supply, and so forth.
(4) Firms have easy entry and exit in the long-run.

What is the implication of the presence many buyers and many sellers in a competitive market?
Buyers and sellers cannot influence the price and they are price takers.

Why does a competitive firm face a horizontal demand curve?
When the equilibrium price has been established, a single perfectly competitive firm faces a horizontal (flat, perfectly elastic) demand curve at the equilibrium price In short, the firm takes the equilibrium price as given—hence the firm is a price taker—and sells all quantities of output at this price.

The marginal revenue curve for the perfectly competitive firm is the same as its demand curve. Why?
In competitive firm price is given for a competitive firm. By selling additional unit it gets MR equal to price i.e., price is equal to marginal revenue (P = MR). A demand curve plots price against quantity, whereas a marginal revenue curve plots marginal revenue against quantity. If price equals marginal revenue, then the demand curve and marginal revenue curve are the same.

What is the profit maximization rule for a short-run competitive firm? How much it produces to maximize profit?
Profit maximization rule: MR = MC. It produces an amount of output for which MR = MC.

Why is a short-run perfectly competitive firm resource allocative efficient?
The perfectly competitive firm is resource allocative efficient as it produces at P = MC.

What are the possible outcomes of a short-run competitive firm?
Outcomes:

(a) Firm can make economic profit (b) firm can make normal profit (c) firm may incur economic loss.

Which curve represents the short-run supply curve of a competitive firm?
Perfectly competitive firm's short-run supply curve is the part of its marginal cost curve above its average variable cost curve,

What is the equilibrium condition of a long-run competitive firm? Why does a long-run competitive firm make only normal profit?
Long-run equilibrium condition: P = MC = SRAC = LRAC. A long-run competitive firm makes normal profit because of the property, free entry and exit in the long-run.

What does the long-run equilibrium condition of competitive firm imply?
No competitive firm has an incentive to (a) enter or exit the industry (b) produce more or less output and (c) change their plant size.